

Master of Science in Knowledge and Technology Management



THE HONG KONG
POLYTECHNIC UNIVERSITY

香港理工大學

Department of Industrial
& Systems Engineering

Managing Knowledge (ISE 542)

*Lecture 3: Knowledge Processes (1): Knowledge
Acquisition, Knowledge Generation and Innovation*



Learning Outcome

- To understand different knowledge processes
- To familiarize knowledge acquisition, knowledge generation and innovation processes and practices
- Differentiate between innovation, knowledge acquisition and knowledge generation



The Knowledge Processes

- Knowledge processes are what individual and collective use to produce, transmit, acquire, store and use knowledge.
- Knowledge management deals with knowledge processes and a knowledge environment.
- There are a number of primary knowledge processes which are:
 - knowledge production (or creation or generation)
 - knowledge acquisition
 - knowledge sharing
 - knowledge retention
 - knowledge validation
 - knowledge transfer (or diffusion).

Input

Data &
Information

Data/Info/Knowledge

Existing Knowledge

Unstructured
Knowledge

Data/Software?
Activity
(Process)
Uses Knowledge

Consumes data/information to make decision

Output

Final Product

e.g. Innovation, share, reuse,
organization learning, etc.

New Knowledge

e.g. Validation processes, peer
review, experiments, etc

Validated
Knowledge

e.g. R&D, research, codification, etc.

Rules, Lessons,
Procedures



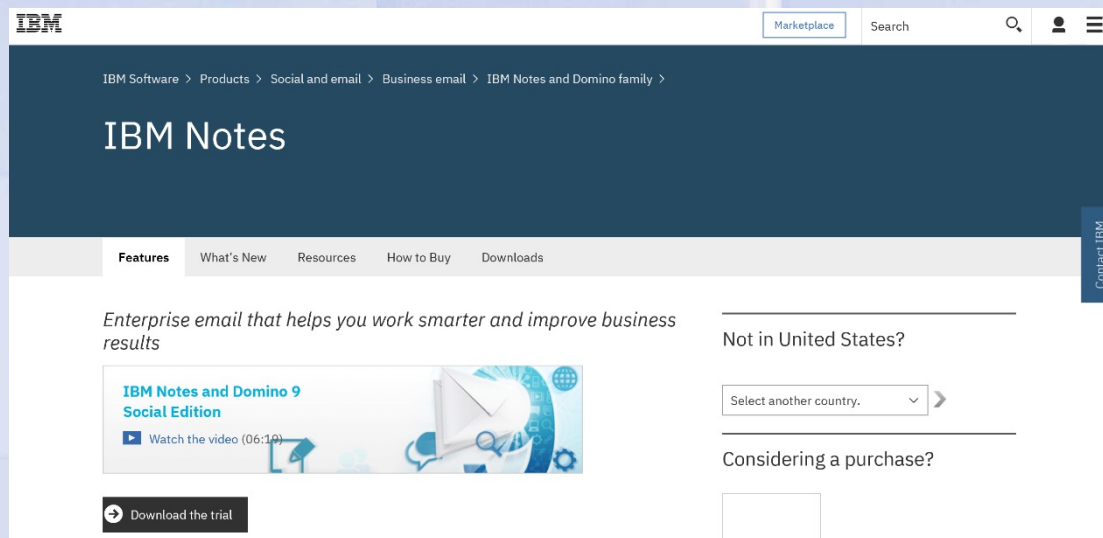
Knowledge Acquisition

- **Knowledge acquisition** presupposes knowledge already exists and that is desired to capture for the benefit of the acquirer.
- Common Practices for knowledge acquisition include:
 - A company might like to capture the knowledge of another firm by acquiring the firm, hiring employees from that firm
 - Reverse engineering one of their products
 - Video/audio recording a person at work
 - Reconstructing the knowledge by examining papers and articles published by the firm because it is perceived that there is important knowledge to be acquired.



Case Example A

- For example, IBM spent USD 3.5 billion to purchase Lotus in 1995 which was 40 times the book value of Lotus of USD250 million. The premium paid by IBM represent their appraisal of Lotus's unique knowledge in collaborative software applications such as Lotus Notes.



Knowledge Production

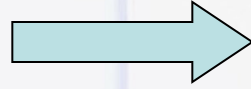
- **Knowledge production** is a process that creates new knowledge through the reconstruction of older concepts and invention of new ones.
- Knowledge is constructed from concepts that already have through further observation of events.
- **Knowledge claim** is a set of rules believed to be true.
- All knowledge creation (personal, group, organization and global) occurs through learning.
- New concepts can be formed through a deep interaction among people in environments that have the conditions to enable knowledge production or creation.





Individual

Learning

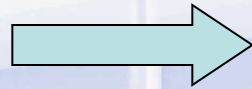


**Personal knowledge
(Individual knowledge claim)**



Collective knowledge
created from group

Collaboration

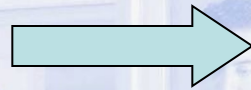


**Group knowledge
(Group knowledge claim)**

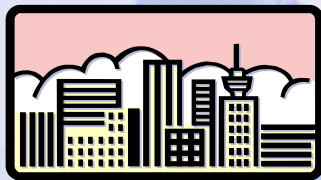


Peer groups

Validation

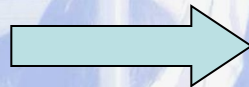


**Organizational knowledge
(Organizational knowledge claim)**



Peer groups from different
organizations

World learn



**Global
knowledge**



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

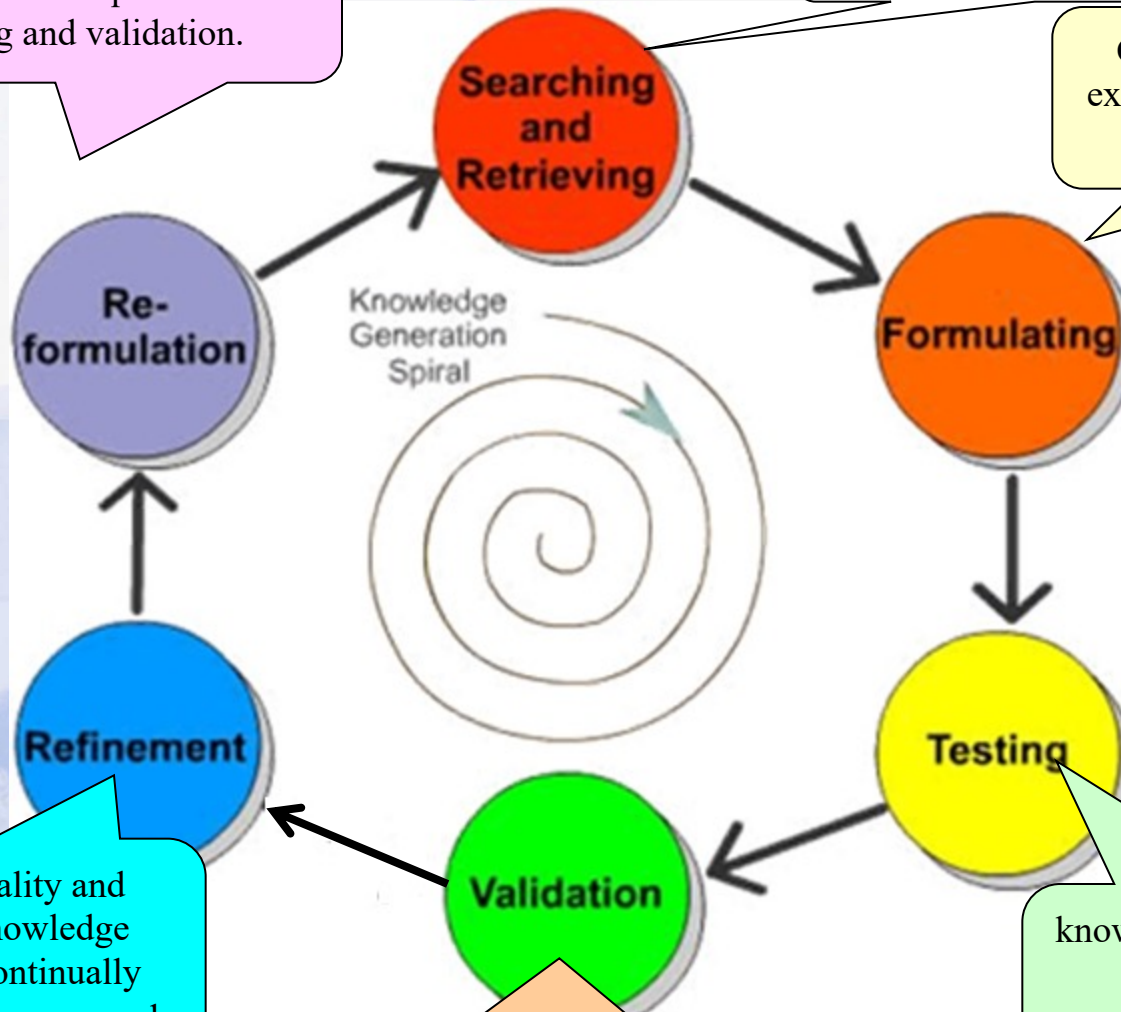
Department of Industrial & Systems Engineering

Knowledge Production Processes

Refinement leads to reformulating a new knowledge claim which requires a new process of testing and validation.

Searching and retrieving older knowledge claims or concepts

Observing, hypothesizing, experimenting, brainstorming, codification, etc.



increase the quality and power of the knowledge claim through continually tested, improved or removed.

knowledge claim goes through a process of testing with experience, peer review, in a laboratory or in the field.

the value of a cluster of knowledge claims is judged and weighed against by knowledge validation strategy.

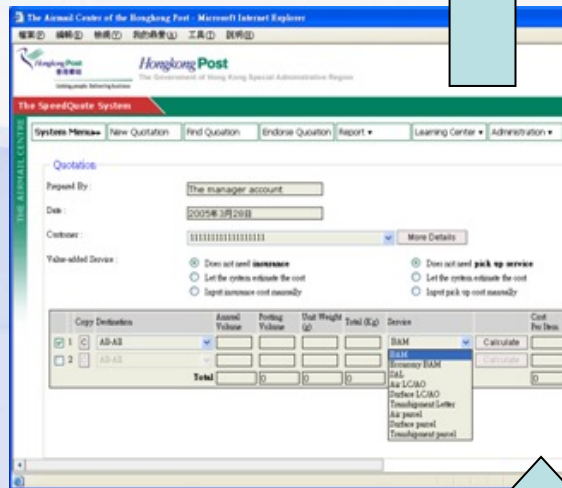


THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

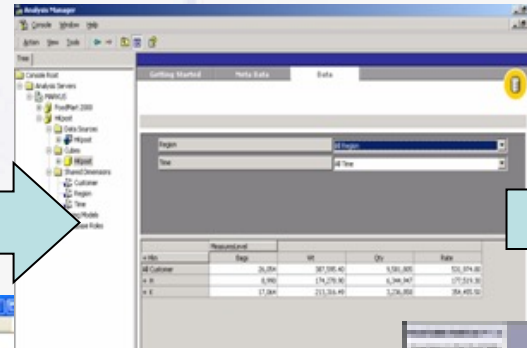
s Engineering

CASE Example B : A Knowledge-based Data Mining System for the Hong Kong Post Office Air Mail Centre

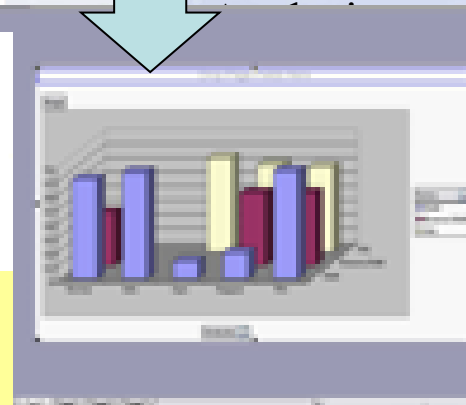
Knowledge-based
Quotation Management



Customer
Relationship
Management



Data Mining
for
Profitability



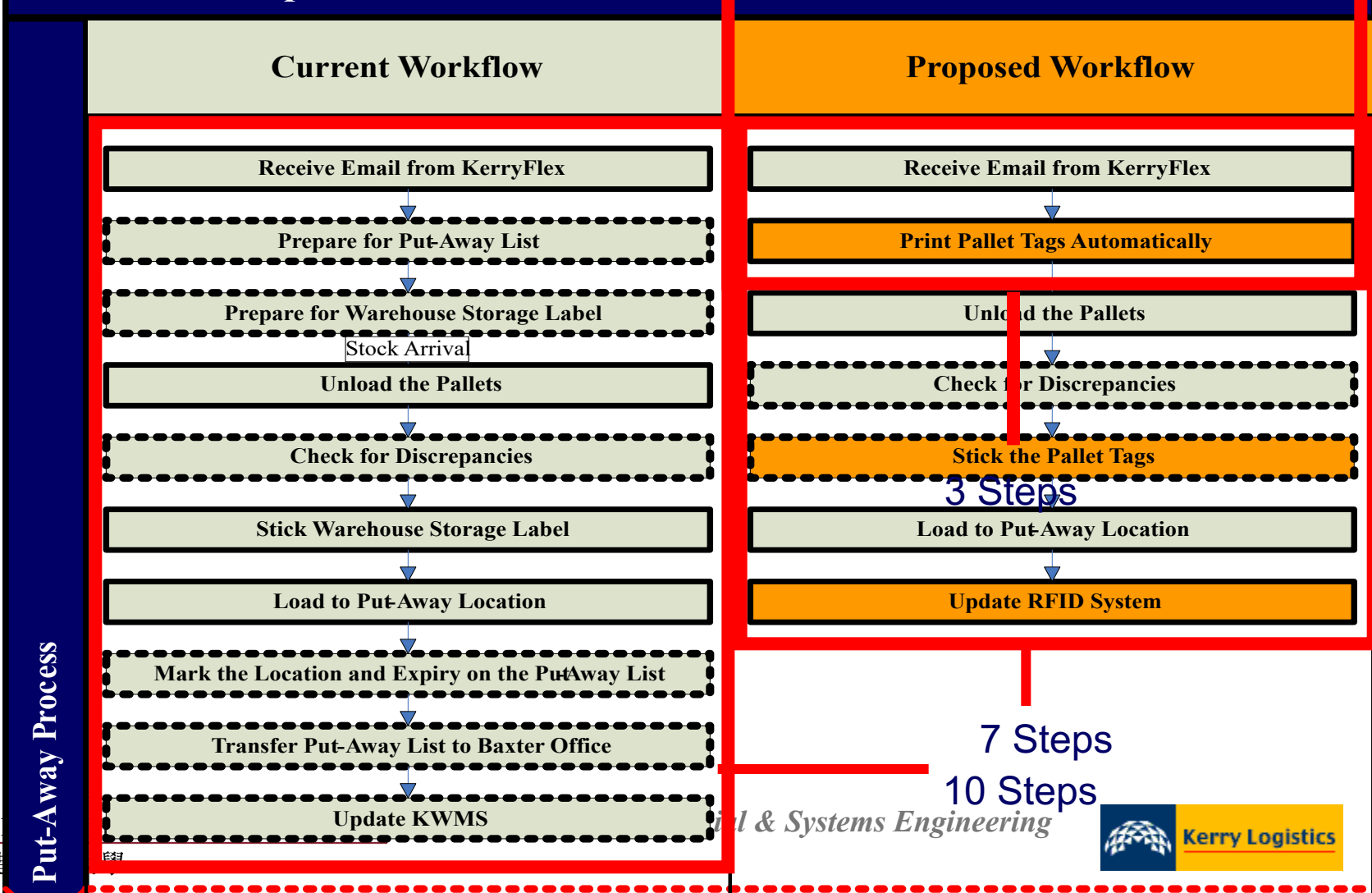
Leveraging Knowledge for
Improving the sales and
profitability of HK Post

Discovery of
Patterns and
Business Rules

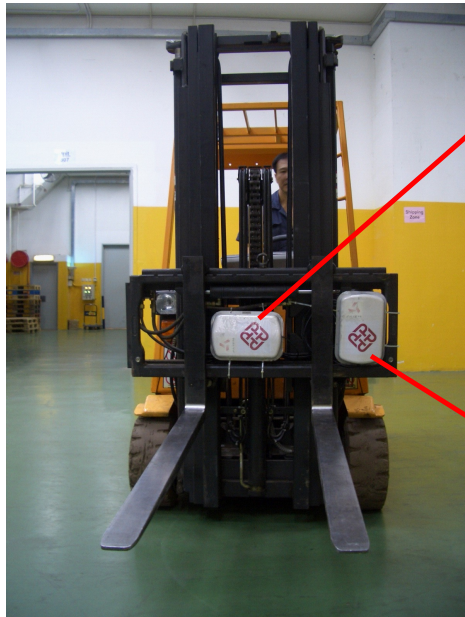


Case study C: Knowledge creation for warehouse management

Workflow Comparison Table



Propose: RFID Forklift



Antenna for scanning the pallet tag

Antenna for scanning the location tag



RFID Tag



RFID Tag

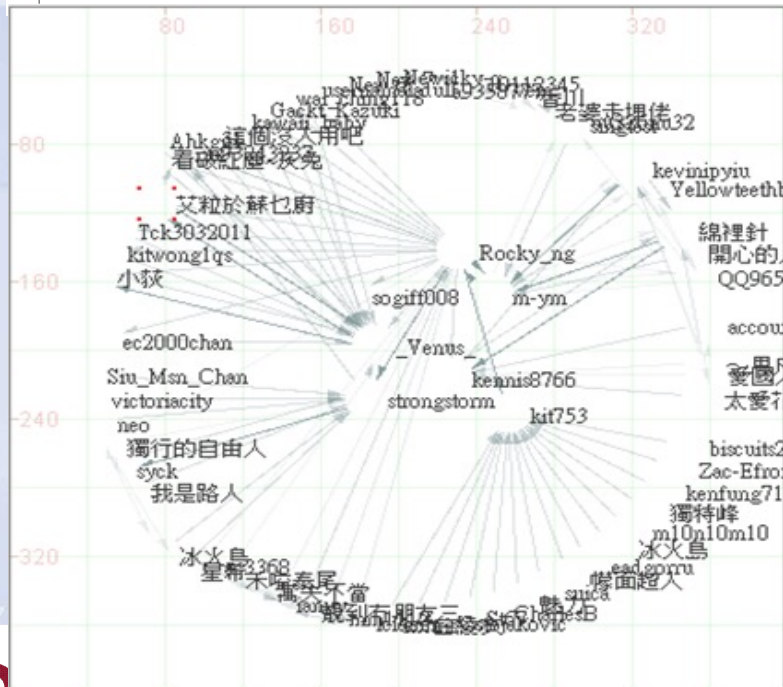
RFID Tag on a cardboard



Case Study D : Knowledge mining to facilitate cyber youth work of Caritas Hong Kong



- A 3 years cyber-youth work supporting program (Cyber Youth Outreach Pilot Scheme)
- Co-operated with Caritas of Hong Kong
- Increasing demand of cyber youth work (The operation mode of social workers)
- Drug-related cases are the main tasks of their work
- Some operation problem of their work
- Weekly placement is conducted



The diagram illustrates the process of N-gram analysis for the sentence "I know the boat P really plays banana boat wake board". The process is shown in a vertical flowchart on the left, with corresponding examples on the right.

Flowchart Steps:

- Input sentence:** The initial sentence to be analyzed.
- Segmentation based on whitespace and punctuations:** The sentence is split into individual words and punctuation marks.
- Segments of sentences:** The resulting list of individual tokens.
- Unigram analysis with checking of English characters:** Each token is analyzed individually.
- Unigrams:** The output of the unigram analysis, which are the individual characters or words.
- N-gram analysis:** The unigrams are combined into groups of size N.
- N-grams:** The final output, which are the groups of N characters or words.

Example Analysis:

An example

Input sentence: "我認識嘅船P 都真係玩下香蕉船 wake board"

Segmentation based on whitespace and punctuations: "我認識嘅船P", "都真係玩下香蕉船", "wake", "board"

Segments of sentences: "我", "認", "識", "嘅", "船", "P", "都", "真", "係", "玩", "下", "香", "蕉", "船", "wake", "board"

Unigram analysis with checking of English characters: (This step is not explicitly shown in the example, but it would involve checking each character for English validity.)

Unigrams: (This step is not explicitly shown in the example, but it would involve breaking down the characters into individual units.)

N-gram analysis: (This step is not explicitly shown in the example, but it would involve combining the unigrams into groups of size N.)

N-grams: (This step is not explicitly shown in the example, but it would involve the final output of the N-gram analysis.)

Bi-gram: "我認", "認識", "識嘅", "嘅船", "船P", "P都", "都真", "真係", "係玩", "玩下", "下香", "香蕉", "蕉船", "船wake", "wake board"

Tri-gram: "我認識", "認識嘅", "識嘅船", "嘅船P", "P都", "都真", "真係", "係玩", "玩下", "下香", "香蕉", "蕉船", "船wake", "wake board"

4-gram: ...

5-gram: ...

6-gram: ...



What is Innovation?

- Innovation defines as the generation of ideas and implementing them to produce value for organizations.
- Innovation includes the concepts of novelty, commercialization and/or implementation.
- Innovation is a process which results in doing new things and/or in doing things in a new way so that value is created.
- *The activities related to* innovation are (to some extent) structured, planned, manageable and output oriented.
- The output of the innovation process should have a new components, product, process or service, etc.
- Should a new idea has not been developed and transformed into a product, process or service, it would not be classified as an innovation.



Examples of Innovation

- Introducing and commercial selling of a new or improved product
- Introducing and commercial using of a new method of production
- Introduction of a new form of business organization
- New uses for existing products
- Develop new markets for existing products
- Develop new channels of distribution



Case Example E

Print edible dye images on potato crisps, e.g.
Pictures, trivia questions, facts, and jokes

- ◆ Searching for a solution in network of institutional and private contacts
- ◆ Bakery in Italy
- ◆ Run by a university professor who manufacture baking equipment.
- ◆ Come up with a way of printing edible images on cakes

Benefits:

- ◆ Cut the cost of development
- ◆ Reduced the concept to market time



Company: Procter and Gamble
Brand: Pringles
Product: Pringles Print (Potato Chips)



In-class-Exercise 1

Classify the following activities into categories: Knowledge Acquisition, Knowledge Production and Innovation:

- I. Using search engine to search for product information
- II. Brainstorming new product ideas
- III. Develop and practically implement a new e-business model in the market
- IV. Improving the efficiency of existing method of doing things by skipping some steps
- V. Hiring consultants for providing technical support of operation

Select the best answer from the following:

	Knowledge Acquisition	Knowledge Production	Innovation
A.	I, V	II, IV	III
B.	I, IV	II	III, V
C.	II, V	I	III, IV
D.	IV, V	I, II	III
E.	V	I, II	III, IV



Innovation Management

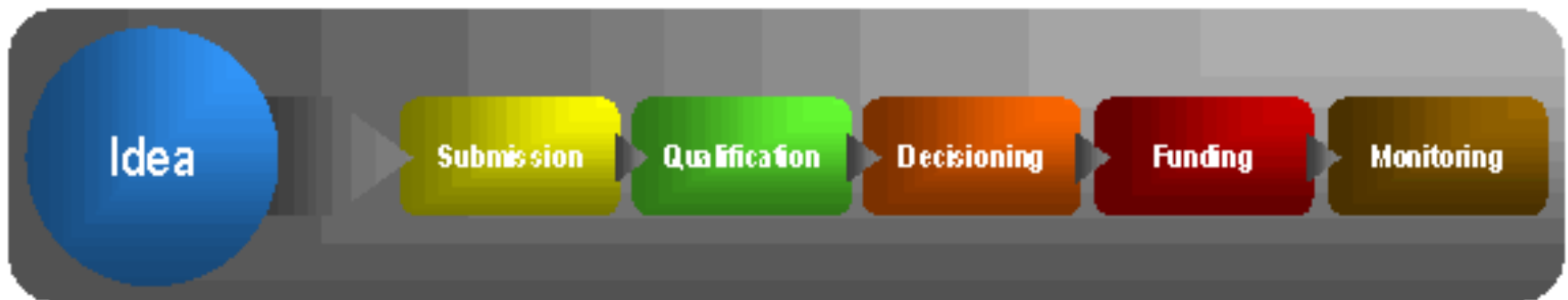
- Innovation management is the discipline of managing processes in innovation.
- Innovation management focuses on allowing the organization to use its creative effort to introduce new ideas, processes or products.



Innovation Process

Idea Generation

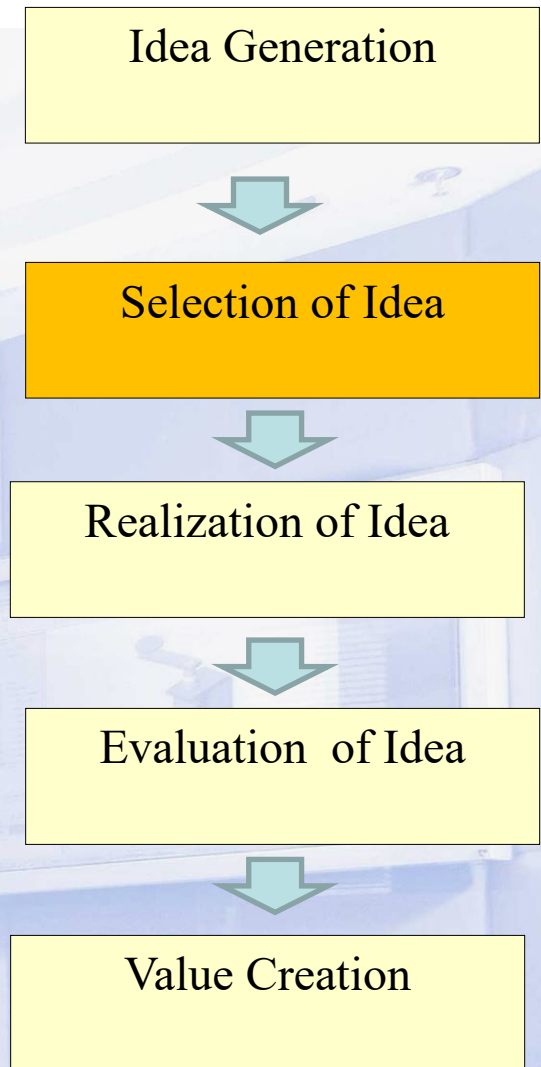
- Once an idea is created, the work begins with the submission phase. This phase consists of 2 main components: the Ideation and the Business Case.
- The Ideation component is intended to help the sponsor clarify and structure the idea for an initiative.
- Ideation process provides an opportunity for the Initiator to test the receptivity within their organisation, and learn from others within the organisation prior to investing in the more labor-intensive development of the business case.



Innovation Process

•Selection of Idea

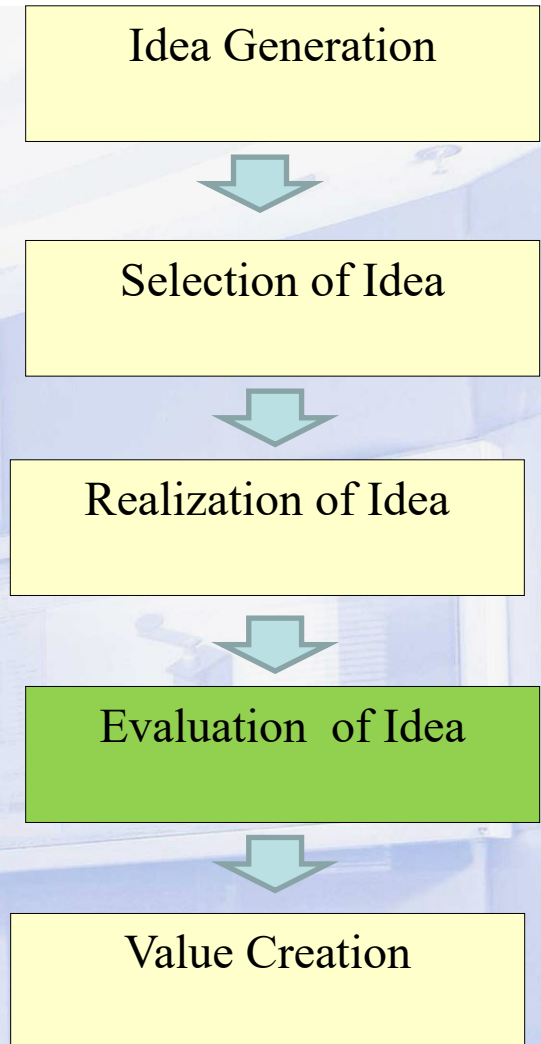
- Selection of ideas is a crucial step in the innovation process
- In the practice, it means that the right information should be provided in the proposal.
- The knowledge about and from the various stakeholders is essential and their needs and concerns should be addressed.
- Based on understanding the strategy and available resources the decision can be made to continue, modify or stop the proposed way forward.



Innovation Process

•Evaluation of Idea

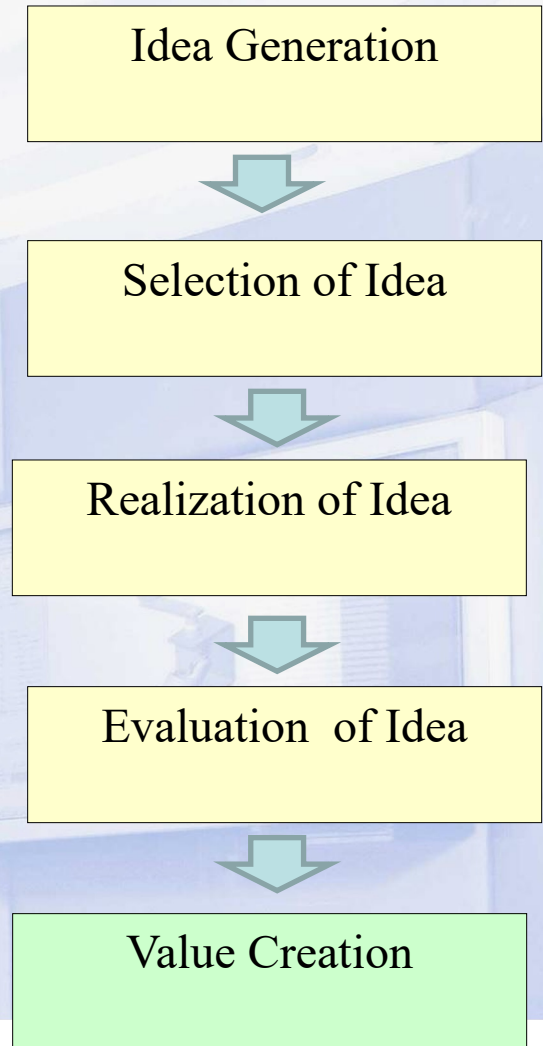
- It is necessary to understand the relation between input and output of the process.
- It is also vital to use the evaluation as feedback for the selection phase.
- During the evaluation, one will validate the assumptions of the proposal. It will also enable capturing the overall lessons learned.



Innovation Process

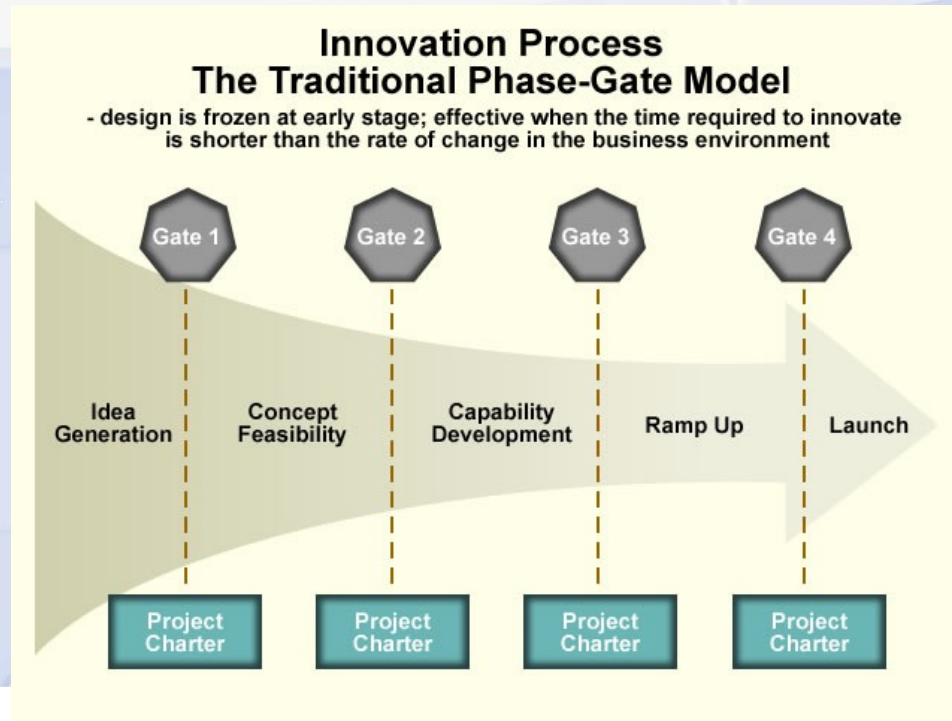
•Value Creation

- Application of the results is an obvious way for creating the value.
- Another way to create value is by dissemination of the results and learning from the innovation process. Here, both single-loop learning, the project outcomes can be used in the application and/or in related or similar projects.
- By reflecting on the innovation process itself the process as a whole could be improved and changes of success will be bigger next time:
- This is double-loop learning.



Development of Business Cases

- During the life-cycle of a start-up company several stages or phases of development are identified. Each phase has different characteristics which is reflected in activities, milestones and subsequently risks. At the end of each phase there is a moment at which the viability of the project, and thus the continuation will have to be re-established.



(Source:

http://www.1000ventures.com/business_guide/im_process_main.html.

Last accessed 5/2/05)



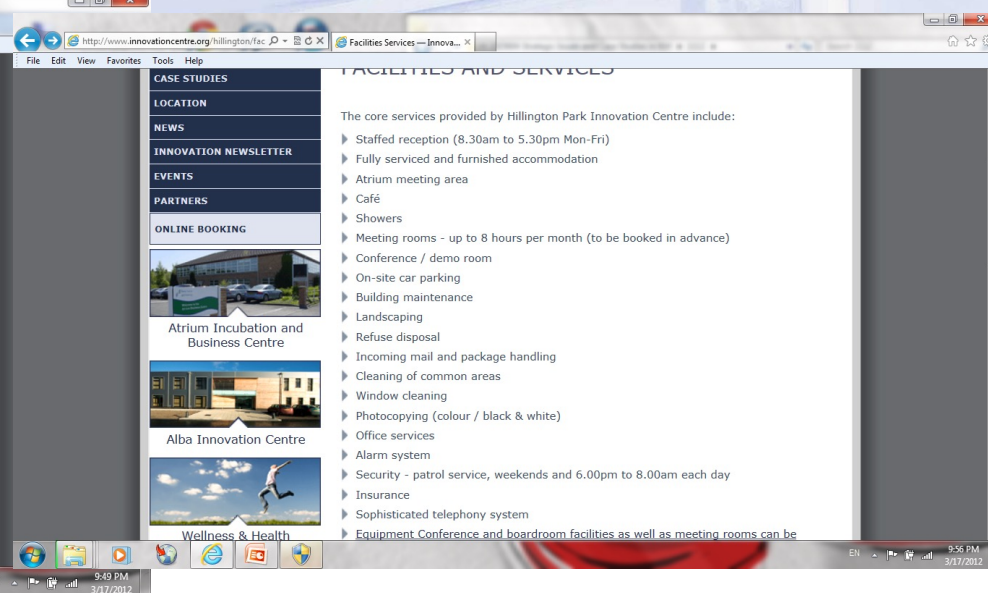
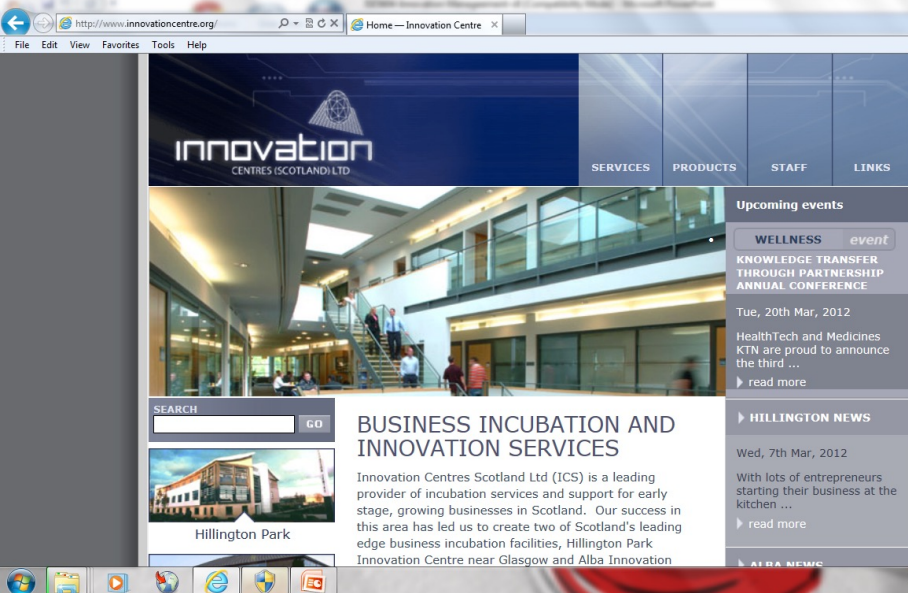
Development of Business Cases

- **Project Description** - Describe what the project does.
- **Business/Product or Service** - Explain the business product or service in simple terms.
- **The Market** - Define the target market for the product or service. Describe the market characteristics, such as growth patterns and key customers.
- **Competition** - Describe the competition, including indirect competitors with alternative products or services. Describe what makes this product or service unique or competitive.
- **Risk/Opportunity** - Describe the most significant risks to the opportunity, including market risk, pricing risk, product risk, and management risk. Describe how these risks can be overcome.
- **Technology Solution** - Describe the target technology solution set that has been defined to date.



Innovation Environment: Innovation Centre

Innovation Centre serves as Innovation Accelerator for enhancing capacity for innovation through hands on engagements. Innovation Centre usually offers a comprehensive set of programs and services to foster innovation.



The Hong Kong Science and Technology Parks Corporation



Hong Kong 香港科技園
Science & Technology Parks

Offer one-stop infrastructural support services to technology-based companies and activities

Science Park provides premises and services for applied R & D activities

Incubation programme nurtures technology and design start-ups

InnoCentre creates and sustains a design cluster

Industrial estates provide land and premises for production



Cyberport

- Provide quality environment and advanced telecommunications and information technology (IT) infrastructure to attract a cluster of world-class companies and professional talents specialising in IT applications and services
- Digital Entertainment Incubation-cum-Training Centre



Hong Kong Design Centre

To promote design as a value-added activity

To raise design standards and foster design-related education

To raise the profile of Hong Kong as an innovative and creative hub in Asia



THE HONG KONG
POLYTECHNIC UNIVERSITY
香港理工大學

Department of Industrial & Systems Engineering